

Computer Netzwerke und verteilte Systeme Summer 2026



Exercise 10: Multicast

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Deadline: The deadline for this exercise is **30.06.2026, 23:55** and in total **13 bonus points** can be achieved.

Bonus points: *Computer Netzwerke und verteilte Systeme* features a bonus point system to acknowledge first-rate performance. Exercise points count as bonus. Throughout the entire course, a bonus of up to a maximum of 0.7 can be obtained. The bonus does not help in passing the exam: You need to pass the exam with a 4.0 or better before any bonus is applied.

Group size: The recommended group size for submissions is 3. You may also submit your exercises as a group of 2 or alone if you do not find a group.

Important: Every theoretical task and subtask require proper sources. Submissions without proper sources will automatically receive 0 points.

For most of the task the slides of the lecture will be enough. For **research** task, use the materials provided in the lecture, or the internet.

Refrain from copying and pasting your answers, as this is considered plagiarism and will have **consequences**, as will using AI tools to create your answers. Instead, explain your answers in your own words and cite your sources!

Write down your answers in a legibly and unambiguously way, either by creating formatted documents (LaTeX, Word, typst, etc.) or by handwriting (**ensure that your handwriting is legible**).

Refrain from using colors that create less contrast between the text and the background. For marking tasks you may use blue and orange.

Task 1: Group Exercise: Multicast Theory

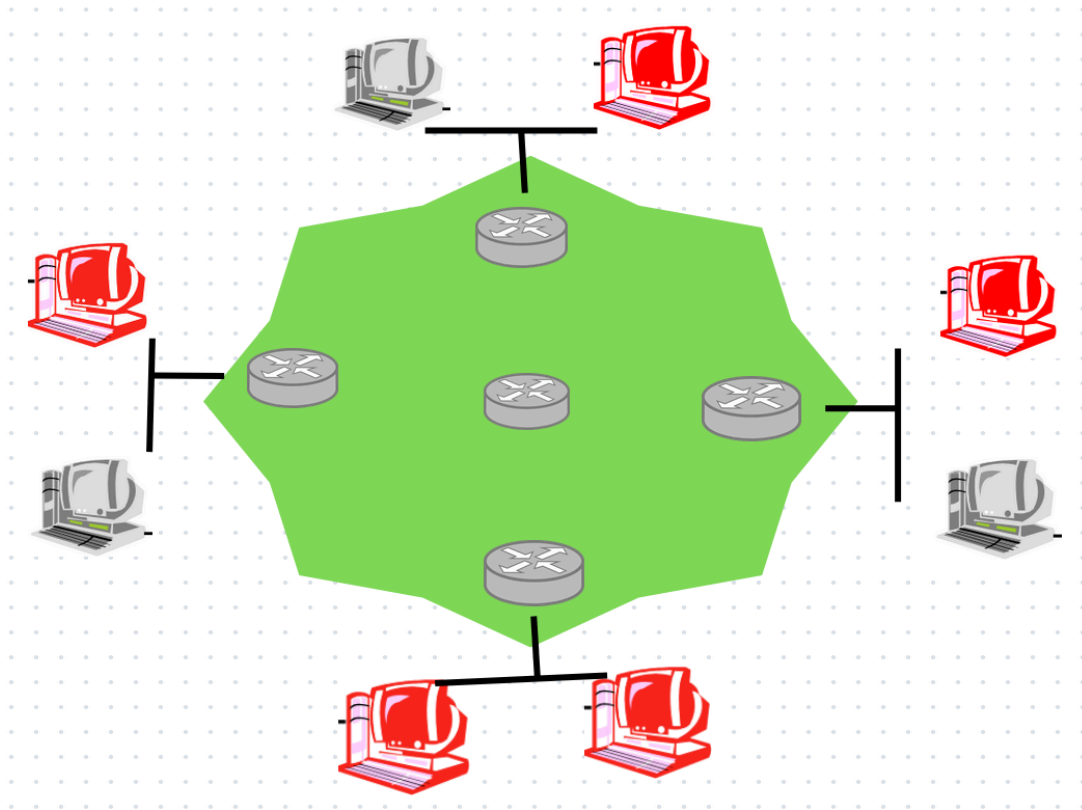
- (a) State three different multicast applications.
- (b) Name three approaches to realize multicasting.
- (c) State three approaches for routing and delivery for multicast.
- (d) Provide two reasons why Steiner trees are not used in practice for multicast routing.
- (e) Which protocol is used for joining a local Multicast Group? Which actions do Host and Router perform?
- (f) What IP address block is reserved for multicast (both IPv4 and IPv6)?
- (g) Briefly describe how ICMP error messages are generated for multicast datagrams.
- (h) IPv4 and IPv6 multicast addresses differ structurally. Explain the role of the “scope” fields in IPv6 multicast addresses.

Task 2: Group Exercise: Multicasts Practice

The client on the left is sending packages via router in the center to the clients marked in red. Draw the following paths:

- (a) Multicast via unicast

- (b) Network multicast
- (c) Application layer multicast



Task 3: Group Exercise: Core Based Tree

Consider the Core Based Tree (CBT) in Figure 5. The center of the CBT is A and the members of the receiver group are E, F, G, H and J.

Show the path through the network that a message takes from sender J to the four remaining receivers. Also show the path that J's message would take if it were not a member. Use Shortest Path First.

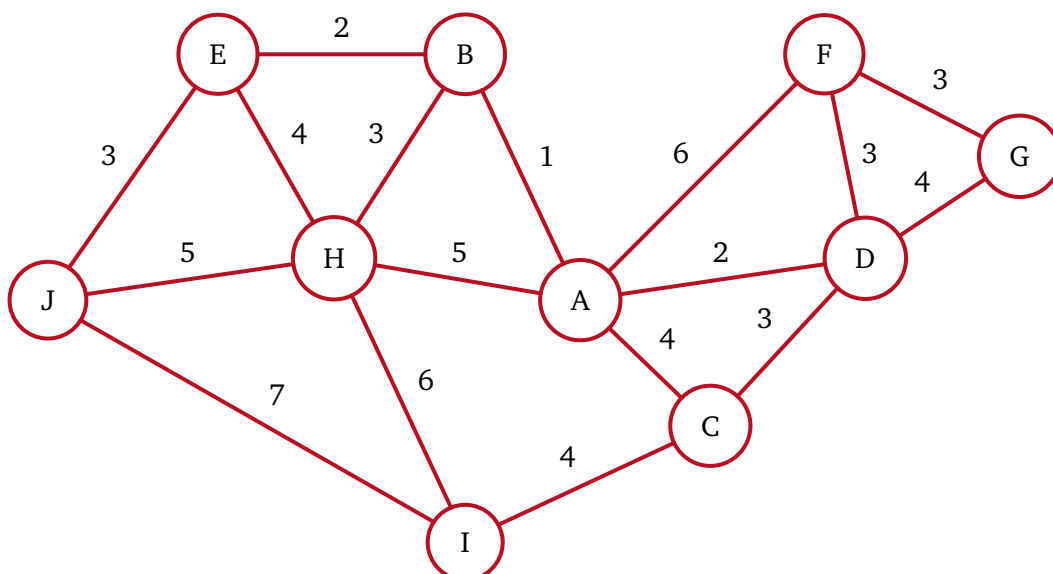


Figure 2: Topology for task 3

Task 4: Theory Questions Multicast (9 P)

- (a) In which situations is multicast preferable to sending separate unicast transmissions, and what advantages does it provide? (1 P)
- (b) How do the three main approaches to multicast work? (1 P)
- (c) What difficulties arise from translating IP multicast group addresses into Ethernet multicast MAC addresses? (2 P)
- (d) If a host connected to a LAN does not respond to an IGMP query, what consequences can this have for group membership detection and multicast forwarding? (2 P)
- (e) Which of the following IP addresses are valid IPv4 Multicast addresses: 192.168.1.10, 224.0.0.0, 224.1.2.3, 226.17.30.197, 240.0.0.1 (1 P)
- (f) For a livestream with 10000 viewers would you rather use *multicast via unicast* or *network multicast*? Explain which links in the network bear more load with *multicast via unicast* and explain why *network multicast* can be more efficient. (2 P)

Attention: Submissions without references will not be graded! Please add sources.

Task 5: Multicast in practice (4 P)

Research and describe a real-world application in which multicast is used (e.g., IPTV, CDN, financial markets). How is multicast implemented there and how does it deal with packet loss?

Attention: Submissions without references will not be graded! Please add sources.